

ISA 634: Analytics Solution Deployment

Department of Information Systems & Analytics

Farmer School of Business

Miami University

Outline

- 1 General, Standard and Matrix Form of Linear Programming Problem**
- 2 Another LP Example: Wyndor Glass Problem
- 3 Assumptions of Linear Programming Problems
- 4 Recap

The General Form of Linear Programming Problems

Variables Def: Let $x_1, x_2, \dots, x_n$ be continuous decision variables

Objective Function: $\max \text{ (or min) } c_1x_1 + c_2x_2 + \dots + c_nx_n$

Constraints:

subject to $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$

$\vdots$

$a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kn}x_n = b_k$

$\vdots$

$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \geq b_m$

$\vdots$

x_i can be ≤ 0 , ≥ 0 , or *free* for $i = 1, \dots, n$

The General Form of Linear Programming Problems

Variables Def: Let $x_1, x_2, \dots, x_n$ be continuous decision variables

Objective Function: $\max \text{ (or min) } c_1x_1 + c_2x_2 + \dots + c_nx_n$

Constraints:

subject to $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$

$\vdots$

$a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kn}x_n = b_k$

$\vdots$

$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \geq b_m$

$\vdots$

x_i can be ≤ 0 , ≥ 0 , or *free* for $i = 1, \dots, n$

The General Form of Linear Programming Problems

Variables Def: Let $x_1, x_2, \dots, x_n$ be continuous decision variables

Objective Function: $\max \text{ (or min) } c_1x_1 + c_2x_2 + \dots + c_nx_n$

Constraints:

subject to $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$

$\vdots$

$a_{k1}x_1 + a_{k2}x_2 + \dots + a_{kn}x_n = b_k$

$\vdots$

$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \geq b_m$

$\vdots$

x_i can be ≤ 0 , ≥ 0 , or *free* for $i = 1, \dots, n$

The *Standard Form* of Linear Programming Problems

Variables Def: Let $x_1, x_2, \dots, x_n$ be continuous decision variables

Objective Function:

$$\min \sum_j^n c_j x_j$$

Constraints:

$$\text{s.t.} \quad \sum_j^n a_{ij} x_j \geq b_i, \quad i = 1, \dots, m$$

$$x_j \geq 0, \text{ for } j = 1, \dots, n$$

- The above representation of an LP is one example of a **Standard Form** of an LP

The *Standard Form* of Linear Programming Problems

Variables Def: Let $x_1, x_2, \dots, x_n$ be continuous decision variables

Objective Function:

$$\min \sum_j^n c_j x_j$$

Constraints:

$$\text{s.t.} \quad \sum_j^n a_{ij} x_j \geq b_i, \quad i = 1, \dots, m$$

$$x_j \geq 0, \text{ for } j = 1, \dots, n$$

- The above representation of an LP is one example of a **Standard Form** of an LP

The *Standard Form* of Linear Programming Problems

Variables Def: Let $x_1, x_2, \dots, x_n$ be continuous decision variables

Objective Function:

$$\min \sum_j^n c_j x_j$$

Constraints:

$$\text{s.t.} \quad \sum_j^n a_{ij} x_j \geq b_i, \quad i = 1, \dots, m$$

$$x_j \geq 0, \text{ for } j = 1, \dots, n$$

- The above representation of an LP is one example of a **Standard Form** of an LP

The *Standard Form* of Linear Programming Problems

- Q1** Why are we only considering “min” in the formulation above? Ans: can convert max to min.
- Q2** What happened to the “ $\geq$ ” and “=” in the constraints above, and the “ $\leq$ ” and “*free*” variable restrictions? Ans: can always convert between $\leq$ and “*free*” and “ $\geq$ ”.

Representing Variables and Parameters in Matrix & Vector Form

Recall the general form of an LP from above:

Objective Function:

$$\min \quad c_1x_1 + c_2x_2 \dots + c_nx_n$$

Constraints:

$$\text{s.t. } a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n (\leq \text{ or } \geq \text{ or } =) b_i, \quad i = 1, \dots, m$$

$$x_j \text{ can be } \leq 0, \geq 0, \text{ or free for } i = 1, \dots, n$$

Define:

$$\mathbf{c}^T = [c_1, \dots, c_n] \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

The *Matrix Form* of Linear Programming Problems

Variables Def: Let $\mathbf{x}$ be a *vector* of n continuous decision variables

Objective Function:

$$\min \quad \mathbf{c}^T \mathbf{x}$$

$$\text{s.t. } \mathbf{Ax} \ (\leq \text{ or } \geq \text{ or } =) \ \mathbf{b}$$

Constraints:

$$\mathbf{x} \ (\leq \text{ or } \geq) \ \mathbf{0} \text{ or } \textit{free}$$

$$\mathbf{c}^T = [c_1, \dots, c_n] \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \quad \mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

A couple relevant matrix operations

- Transpose of a vector $\mathbf{v}$
- Vector multiplication $\mathbf{c}^T \mathbf{x}$
- Multiplication $\mathbf{Ax}$

Diet Problem Revisited

Recall The Diet Problem:

My diet requires that all the food I eat come from one of the four “basic food groups” (chocolate cake, ice cream, soda, and cheesecake). At present, the following four foods are available for consumption: brownies, chocolate ice cream, cola, and pineapple cheesecake. Each brownie costs 50 cents, each scoop of chocolate ice cream costs 20 cents, each bottle of cola costs 30 cents, and each piece of pineapple cheesecake costs 80 cents. Each day, I must ingest at least 500 calories, 6 oz of chocolate, 10 oz of sugar, and 8 oz of fat. The nutritional content per unit of each food is shown in the table below. Formulate a linear programming model that can be used to satisfy my daily nutritional requirements at minimum cost.

Nutritional Values for Diet

Type of Food	Calories	Chocolate (Ounces)	Sugar (Ounces)	Fat (Ounces)
Brownie	400	3	2	2
Chocolate ice cream (1 scoop)	200	2	2	4
Cola (1 bottle)	150	0	4	1
Pineapple cheesecake (1 piece)	500	0	4	5

Source: Operations Research Applications and Algorithms - WL Winston (Thomson Learning, 2004) Page 69.

Diet Problem Revisited

Q: What is the $\mathbf{A}$ matrix, decision vector $\mathbf{x}$, rhs vector $\mathbf{b}$, and cost vector $\mathbf{c}^T$ in the Diet Problem?

Solving the Diet Problem Using Excel and R

Please download the posted Excel and R code file for the Diet Problem from Canvas. We'll next work through these together.

A Practice Example

Identify the A matrix and c^T , x , b vectors for the LP below:

$$\begin{aligned} \max \quad & 5x_1 - 3x_2 - x_3 \\ \text{subject to} \quad & x_1 + 2x_2 - 7x_3 - 5 \leq 0 \\ & -3x_1 + x_2 + x_3 - 4 = 0 \\ & 9x_1 + 2x_2 + 4x_3 - 1 \geq 11 \\ & x_2 - x_3 + x_1 \leq 9 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Outline

- 1 General, Standard and Matrix Form of Linear Programming Problem
- 2 Another LP Example: Wyndor Glass Problem**
- 3 Assumptions of Linear Programming Problems
- 4 Recap

Class Exercise (Wyndor Glass Co. Problem)

The Wyndor Glass Co. produces high-quality glass products, including windows and glass doors. Each product will be produced in batches of 20, so the production rate is defined as # batches produced per week. It has three plants where one or both of the products are manufactured (i.e., a product may consume resources of one or more plants). Any combination of production rates that satisfies available production time restrictions is permitted. Given the data table below, formulate an LP that can determine what the production rates should be for the two products in order to maximize Wyndor Glass Co.'s total profit, subject to the limited production capacities available in the three plants.

TABLE 3.1 Data for the Wyndor Glass Co. problem

Plant	Production Time per Batch, Hours		Production Time Available per Week, Hours
	Product		
	1	2	
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	\$3,000	\$5,000	

Source: Hillier, F.S., & Lieberman, G.J. (2001). Introduction to Operations Research, 7th Edition. McGraw-Hill Education, pp. 25-31.

A Mathematical Model Representing the Wyndor Glass Co. Problem

Solve Wyndor Glass Co. Problem Using Excel and R

Please solve the Wyndor Glass Co. problem using Excel and R. You can use the files I gave you for the Diet Problem as a reference.

Class Exercise: LP or not?

Based on the information, we have covered in class so far. Please answer the following Yes/No Questions.

- **Is the “Fence Installation Problem” a linear programming problem?**
- **Is the “Diet Problem” a linear programming problem?**
- **Is the “Wyndor Glass Co. Problem” a linear programming problem?**

Outline

- 1 General, Standard and Matrix Form of Linear Programming Problem
- 2 Another LP Example: Wyndor Glass Problem
- 3 Assumptions of Linear Programming Problems**
- 4 Recap

Important Behind the Scene Assumptions in LP Models (1)

Effects of LP Objective Function Being Linear

- **Proportionality:** contribution to the objective function from each decision variable is proportional to the value of the decision variable.
 - What is the effect of having $x_1 = 4$ compared $x_1 = 1$ in the Windsor Glass Problem?
- **Additivity:** contribution to objective function from any decision variable is independent of the values of the other decision variables.
 - No matter what is the value of x_2 , the production of x_1 will always contribute 3 (in thousands) to the objective function of the previous problem

Important Behind the Scene Assumptions in LP Models (1)

Effects of LP Objective Function Being Linear

- **Proportionality:** contribution to the objective function from each decision variable is proportional to the value of the decision variable.
 - What is the effect of having $x_1 = 4$ compared $x_1 = 1$ in the Windsor Glass Problem?
- **Additivity:** contribution to objective function from any decision variable is independent of the values of the other decision variables.
 - No matter what is the value of x_2 , the production of x_1 will always contribute 3 (in thousands) to the objective function of the previous problem

Important Behind the Scene Assumptions in LP Models (1)

Effects of LP Objective Function Being Linear

- **Proportionality:** contribution to the objective function from each decision variable is proportional to the value of the decision variable.
 - What is the effect of having $x_1 = 4$ compared $x_1 = 1$ in the Windsor Glass Problem?
- **Additivity:** contribution to objective function from any decision variable is independent of the values of the other decision variables.
 - No matter what is the value of x_2 , the production of x_1 will always contribute 3 (in thousands) to the objective function of the previous problem

Important Behind the Scene Assumptions in LP Models (1)

Effects of LP Objective Function Being Linear

- **Proportionality:** contribution to the objective function from each decision variable is proportional to the value of the decision variable.
 - What is the effect of having $x_1 = 4$ compared $x_1 = 1$ in the Windsor Glass Problem?
- **Additivity:** contribution to objective function from any decision variable is independent of the values of the other decision variables.
 - No matter what is the value of x_2 , the production of x_1 will always contribute 3 (in thousands) to the objective function of the previous problem

Important Behind the Scene Assumptions in LP Models (2)

Effects of Constraints Being Linear

- **Proportionality:** contribution of each decision variable to the left-hand side of each constraint is proportional to the value of the variable.
 - In plant 3, the production of 1 batch of product 1 always takes 1.5 hours more than 1 batch of product 2.
 - No Economies of Scale
- **Additivity:** contribution of a decision variable to the left-hand side of a constraint is independent of the values of the other decision variables.
 - No matter what is the value of x_1 , the production of one batch of x_2 will always take 2 hours of labor

Important Behind the Scene Assumptions in LP Models (2)

Effects of Constraints Being Linear

- **Proportionality:** contribution of each decision variable to the left-hand side of each constraint is proportional to the value of the variable.
 - In plant 3, the production of 1 batch of product 1 always takes 1.5hours more than 1 batch of product 2.
 - No Economies of Scale
- **Additivity:** contribution of a decision variable to the left-hand side of a constraint is independent of the values of the other decision variables.
 - No matter what is the value of x_1 , the production of one batch of x_2 will always take 2 hours of labor

Important Behind the Scene Assumptions in LP Models (2)

Effects of Constraints Being Linear

- **Proportionality:** contribution of each decision variable to the left-hand side of each constraint is proportional to the value of the variable.
 - In plant 3, the production of 1 batch of product 1 always takes 1.5 hours more than 1 batch of product 2.
 - No Economies of Scale
- **Additivity:** contribution of a decision variable to the left-hand side of a constraint is independent of the values of the other decision variables.
 - No matter what is the value of x_1 , the production of one batch of x_2 will always take 2 hours of labor

Important Behind the Scene Assumptions in LP Models (2)

Effects of Constraints Being Linear

- **Proportionality:** contribution of each decision variable to the left-hand side of each constraint is proportional to the value of the variable.
 - In plant 3, the production of 1 batch of product 1 always takes 1.5hours more than 1 batch of product 2.
 - No Economies of Scale
- **Additivity:** contribution of a decision variable to the left-hand side of a constraint is independent of the values of the other decision variables.
 - No matter what is the value of x_1 , the production of one batch of x_2 will always take 2 hours of labor

Important Behind the Scene Assumptions in LP Models (2)

Effects of Constraints Being Linear

- **Proportionality:** contribution of each decision variable to the left-hand side of each constraint is proportional to the value of the variable.
 - In plant 3, the production of 1 batch of product 1 always takes 1.5hours more than 1 batch of product 2.
 - No Economies of Scale
- **Additivity:** contribution of a decision variable to the left-hand side of a constraint is independent of the values of the other decision variables.
 - No matter what is the value of x_1 , the production of one batch of x_2 will always take 2 hours of labor

Important Behind the Scene Assumptions in LP Models (3)

Other Assumptions

- **Divisibility:** Each decision variable is allowed to take fractional values.
- **Certainty:** Each parameter is known with certainty (whether we are talking about objective function coefficients, coefficients in the LHS of the constraints, or the constants on the RHS of the constraints)

Important Behind the Scene Assumptions in LP Models (3)

Other Assumptions

- **Divisibility:** Each decision variable is allowed to take fractional values.
- **Certainty:** Each parameter is known with certainty (whether we are talking about objective function coefficients, coefficients in the LHS of the constraints, or the constants on the RHS of the constraints)

Outline

- 1 General, Standard and Matrix Form of Linear Programming Problem
- 2 Another LP Example: Wyndor Glass Problem
- 3 Assumptions of Linear Programming Problems
- 4 Recap**

A Summary of Main Learning Objectives from Today's Class

Learning Objectives

Upon completion of today's class, you should be able to:

- Formulate simple optimization problems
- Solve optimization problems
 - Any problem in Solver (if it converges)
 - Linear programming problems using R
- Explain what makes an optimization model a “linear programming” problem.